

TABLE I
PROPERTIES OF COMMON ALLOYS

Alloy Group	Material	Composition (in %)	Resistivity $10^{-6} \Omega m$	TCR ($10^{-3} \Omega/^{\circ}C$)	Max operating temperature ($^{\circ}C$)
Copper	Constantan	54Cu - 45Ni - 1Mn	0.485	0.2	400
Copper	Nickelin	67Cu - 30Ni - 3Mn	0.4	0.11	300
Copper	Manganin	86Cu - 2Ni - 12Mn	0.442	0.02	300
Silver	N.B.W. 109	82Ag - 10Mn - 8Sn	0.55	0-0.04	
Silver	N.B.W. 139	78Ag - 13Mn - 9Sn	0.61	0-0.08	0-150
Silver	N.B.W. 173	80Ag - 17Mn - 3Sn	0.58	0-0.105	0-200
Nickel-chromium	Nichrome	20Cr - 77/80Ni - 0/2Mn	1.105	0.17	1100/1150
Iron-chromium	CrNiFe 1	70Ni - 20Cr - 8Fe - 2Mn	1.11	0.90	1050/1100
Iron-chromium	CrNiFe 2	63Ni - 15Cr - 20Fe - 2Mn	1.12	0.89	1050/1100
Iron-chromium-aluminium	Kanthal A	72Fe - 20Cr - 5Al - 3Co	1.45	0.06	1300
Iron-chromium-aluminium	Cekas	75Fe - 20Cr - 5Al	1.4	0.04	1300
Iron-chromium-aluminium	Megapyr	65Fe - 30Cr - 5Al	1.4	0.025	1350
Pure metals	Tungsten	100 W (sintered)	0.0553	4.5	1500/1700

TABLE II
FIXED CAPITAL

Production capacity per annum			960,000 pieces
Motive power			25kVA (approx.)
LAND AND BUILDING			
Description			Details
Built-up area			400sqm
Office and stores			50sqm
Assembling and testing			350sqm
Rent payable (₹ per annum)			60,000
MACHINERY AND EQUIPMENT			
Description	Indian/Imported	Quantity	Amount (₹)
Toggle action press ½-ton capacity	Indian	2	20,000
Automatic coil winding machine	Indian	3	85,000
Lead cutting machine grinder	Indian	1	9,000
Lead straightening machine	Indian	1	7,000
Lead tinning machine	Indian	1	50,000
Spot welding machine	Indian	1	15,000
Oven (temp.)	Indian	1	30,000
Stirrer (ball mills)	Indian	3	20,000
LCR - Q meter	Indian	1	20,000
Digital temperature indicator	Indian	1	9,500
4½-digit bench micro-ohm meter	Indian	1	15,500
Dimmer stat (4A)	Indian	1	8,000
4½-digit digital multimeter	Indian	1	10,000
Load tester and brake tester	Indian	1	4,000
Fusing tent machine	Indian	1	5,000
Tension tester	Indian	1	6,000
Marking machine with accessories	Indian	1	5,000
Resistance oil standards	Indian	LS	10,000
Total			329,000
Electrification @ 10% of the above			32,900
Office equipment/furniture			60,000
Jigs/fixtures			18,000
Pre-operative Expenses			20,000
Total fixed capital			459,900

LCD monitors and digital cameras are escalating the demand for wirewound variable resistors.

Growing adoption of circuit-breaking applications is another factor boosting the growth of the wirewound variable resistors market. Use of pure metals and alloys that have high temperature coefficient of resistance is increasing significantly, thereby increasing the demand for wirewound variable resistors, to prevent excess electric current transfer.

Due to an abundance of opportunities, a number of players are entering the market. And, existing players are investing heavily in research and development to improve the functionalities of wirewound variable resistors.

Production of wirewound variable resistors is increasing due to low technology barriers and an abundance of raw materials in the market. However, adoption of alternative technologies (such as chip resistors) and macroeconomic conditions (such as economic slowdown and global recession) may hamper the growth of the wirewound variable resistors market.

Precision resistors are expected to grow rapidly owing to the number of benefits provided by these, such as better short-term and long-term stability, faster frequency response, low noise and high performance. Manufacturers are focusing on providing enhanced precision resistors.

Manufacturing process

The wirewound resistor manufacturing process varies widely. The manufacturing process and choice of materials to be used depend on the way the resistor will be used in a circuit. However, all wirewound resistors are made by winding the resistance wire in a spiral around a non-conductive core.

The resistance wire is usually a nickel-chromium alloy, and the core is often ceramic or fibre-glass. A coating such as vitreous enamel is used for protection. Spiral winding has capacitive and inductive effects that make it unsuitable for applications higher than 50kHz. Often, other winding techniques are used to reduce the undesired high frequency effects.

For high tolerance requirements, resistance value is measured to determine the exact cut-to-length of the wire.